

XUE Boyang 薛博阳

Email: byxue@se.cuhk.edu.hk; beyondhsueh1024@gmail.com.

Personal Website: <https://amourwaltz.github.io>.

Phone No.: (86) 189 9270 5010 (WeChat); (852) 6024 3345.

Supervisor: Prof. Kam-Fai Wong (ACL Fellow).

Address: CUHK, Shatin, N.T., Hong Kong SAR, P.R. China.

Research Interests: LLM/Agent Post-training, Context Engineering, Trustworthy AI, Speech Recognition, etc.



Education

The Chinese University of Hong Kong, Hong Kong SAR, China. Aug. 2021 - Jul. 2026 (Expected)

Ph.D. Candidate in Department of System Engineering and Engineering Management.

Huazhong University of Science and Technology, Wuhan, China. Sep. 2016 - Jun. 2020

B.Eng. in School of Artificial Intelligence and Automation, Excellent Class.

Research Experiences

- ◆ Visiting Researcher at Rycolab, AI Center, ETH Zurich. Jan. 2026 - Jun. 2026
Research Topic: Reliable LLM. Advisor: [Ryan Cotterell](#)
Project: We propose to apply Information Bottleneck theory to analyze the context of LLM agent and optimize the context to maximize the utility of context information to improve performance of agents.
- ◆ Visiting Researcher at Web Intelligence Group, AI Center, UCL. Aug. 2025 - Oct. 2025
Research Topic: Context Engineering, Search Agent. Advisor: [Prof. Aldo Lipani](#) and [Prof. Emine Yilmaz](#)
Project: The contexts of multi-turn search agents are lengthy and complex. To mitigate context interference of search agents to improve the reliability and efficiency [1], we first identify the context interference effects of different parts of contexts. We then employ several context refinement baselines and propose a distill-based context refiner. Finally, we propose to incorporate context refinement into search agent RL training pipelines (CRRL), which yield further improvements of search agents. The related work [1] accepted in ACL 2026. We are now extending the work to agent memory, aiming to explore the best configuration of context to generate agents' desired behaviors.
- ◆ Research Intern at Noah's Ark Lab, HUAWEI. Aug. 2024 - Jun. 2025
Research Topic: Self-Training, Reliable Reasoning LLM. Advisor: [Dr. Fei Mi](#) and [Dr. Qi Zhu](#)
Project: LLMs tend to fabricate unreliable responses when confronted with problems beyond their capability, we define the reliable LLM on math reasoning tasks and propose a **ReliableMath** benchmark [11] with both solvable and high-quality unsolvable math problems with human check. We evaluate the benchmark on various LLMs and present several valuable findings. Finally, we propose an alignment method to enable LLMs to critically think and identify their capability boundary on both in-domain and out-of-domain tasks. The related work [11] is under review of TACL. Another work [12] by employing difficulty-aware self-training to improve reliability of LLMs on reasoning tasks is in progress.
- ◆ Ph.D. Student at MoE Lab of High Confidence Software Technologies, CUHK. Sep. 2022 - Present
Research Topic: LLM Reliability, Multilingualism, etc. Advisor: [Prof. Kam-Fai Wong](#)
Project 1: LLMs struggle to accurately express the factual knowledge they possess, especially in cases where the LLMs' knowledge boundaries are ambiguous, we propose **UAlign** [2] framework to leverage two uncertainty estimations of confidence and entropy to represent knowledge boundaries, and then explicitly incorporate uncertainty into prompts for LLMs to align with factuality using PPO. **UAlign** can enhance LLMs' capacities to

of knowledge expression to confidently answer known questions and refuse unknown questions. The related work [2] has been accepted in ACL 2025.

Project 2: Current LLM confidence estimations in languages other than English remain underexplored, we propose a comprehensive investigation of multilingual confidence estimations (**MlingConf**) [3] on both language-agnostic (LA) and language-specific (LS) tasks for LLMs. Experiments present valuable findings regarding language dominance effects of English on LA, and query-related languages on LS respectively. We finally propose a Native-Tone Prompting (NTP) Strategy which first prompts LLMs to employ query-related language to answer the to enhance reliability. The related work [3] has been accepted in Findings of ACL 2025.

Collaborated Projects: Develop a more parameter-efficient LoRA module using partial rotation [8]; Develop a benchmark to evaluate LLMs' ability to plan and execute multiple APIs from various sources to complete the user's task [7]; Propose to utilize confidence estimations to decompose complex questions with a LLM self-divide-and-conquer framework [6]; Complete a survey about reasoning economy on LLMs [13].

◆ **Research Intern at Noah's Ark Lab, HUAWEI.**

Dec. 2022 - Jul. 2023

Research Topic: Dialogue Systems, Retrieval-Augmented Generation.

Advisor: [Dr. Weichao Wang](#)

Project: LLMs based retrieval-augmented dialogue systems are prone to generate factually inconsistent responses that fail to accurately express the provided factual knowledge, we address factual inconsistency [4] with **K-Dial**, which explicitly extends FFNs in Transformers to enhance factual knowledge expressions, as well as the implicit reinforcement learning for factual consistency (**RLFC**) to adjust FFNs' expressions by aligning with factual knowledge using PPO. The combined use of **K-Dial** and **RLFC** can significantly enhance the factual consistency of dialogue systems. The related work [4] has been accepted to Findings of EMNLP 2023.

Collaborated Projects: We propose a multi-task framework function as a universal retriever aiming to attention to the relevant context from the long dialogue and retrieve suitable candidates [9].

◆ **Research Assistant and Ph.D. Student at HCCL Lab, CUHK.**

Sep. 2020 - Aug. 2022

Research Topic: Bayesian Learning, Speech Recognition.

Advisor: [Prof. Xunying Liu](#)

Project: Large-scale language models with highly complex architectures using point estimated parameters fail to account for model uncertainty and is prone to overfitting and poor generalization given limited training data. We first investigate to apply Bayesian learning based adaptation on low-resource tasks, which has been accepted to ICASSP 2021 [5]. Then we systematically propose to apply Bayesian learning framework to account for uncertainty modeling to LMs [10]. Different types of uncertainty modeling can be applied to model parameters, model architectures, and hidden representations. Efficient variational inference approach to estimate the latent posterior distribution is adopted. The related work [10] has been accepted to IEEE/ACM TASLP.

◆ **Research Intern at IMDS Lab, HUST.**

Sep. 2019 - Jun. 2020

Research Topic: Medical Deep Learning, Patient-specific Diagnosis.

Advisor: [Prof. Ye Yuan](#)

◆ **Leader of Intelligent Robotics Team, Qi Ming College, HUST.**

Nov. 2017 - Aug. 2019

Projects: Served as programmer for embedded systems for intelligent cars.

Advisor: [Prof. Dingxin He](#)

Awards & Honors

- ◆ CUHK Overseas Research Attachment Programme (ORAP) Award.
- ◆ Excellent Bachelor Thesis, Excellent Student Cadre, Excellent Graduate Student.
- ◆ Academic Scholarship of *Huazhong University of Science and Technology* in 2018-19 academic year.
- ◆ National Grand and First Prize in the 14th and 13th NXP Cup National University Intelligent Car Race.
- ◆ Second Prize in the 19th Asia and Pacific Mathematical Contest in Modeling.

Extracurricular Activities

- ◆ Serving as a reviewer of *ACL*, *EMNLP*, *NAACL*, *COLING*, etc.
- ◆ Serving as a co-founder and organizer of a [NLP Academic Exchange Platform \(NICE\)](#).

- ◆ Serving as a part-time author to write academic blogs on WeChat officials and [Zhihu](#).
- ◆ Served as the secretary of the Youth League in HUST (2017 - 2020).

Selected Publications

Conference

- [1] **Boyang Xue**, Bin Wu, Shuofei Qiao, et al. “Mitigating Context Interference for Reliable and Efficient Search Agent”, ACL 2026.
- [2] **Boyang Xue**, Fei Mi, Qi Zhu, et al. “UAlign: Leveraging Uncertainty Estimations for Factuality Alignment on Large Language Models”, ACL 2025.
- [3] **Boyang Xue**, Hongru Wang, Rui Wang, et al. “MlingConf: A Comprehensive Study of Multilingual Confidence Estimation on Large Language Models”, Findings of ACL 2025.
- [4] **Boyang Xue**, Weichao Wang, Hongru Wang, et al. “Improving Factual Consistency for Knowledge-Grounded Dialogue Systems via Knowledge Enhancement and Alignment”, Findings of EMNLP 2023.
- [5] **Boyang Xue**, Jianwei Yu, Junhao Xu, et al, “Bayesian Transformer Language Models for Speech Recognition”, ICASSP 2021.
- [6] Hongru Wang*, **Boyang Xue***, et al. “Self-DC: When to retrieve and When to generate? Self Divide-and-Conquer for Compositional Unknown Questions”, NAACL 2025 (Oral).
- [7] Hongru Wang, Rui Wang, **Boyang Xue**, et al. “AppBench: Planning of Multiple APIs from Various APPs for Complex User Instruction”, EMNLP 2024.
- [8] Sheng Wang, **Boyang Xue**, Jiacheng Ye, et al. “PRoLoRA: Partial Rotation Empowers More Parameter-Efficient LoRA”, ACL 2024.
- [9] Hongru Wang, **Boyang Xue**, Baohang Zhou, et al. “UniRetriever: Multi-task Candidates Selection for Various Context-Adaptive Conversational Retrieval”, COLING 2024.

Journal

- [10] **Boyang Xue**, Shoukang Hu, Junhao Xu, et al. “Bayesian Neural Network Language Models for Speech Recognition”, TASLP (SCI & JCR Q1), 2022.
- [11] **Boyang Xue**, Qi Zhu, Rui Wang, et al. “ReliableMath: Benchmark of Reliable Mathematical Reasoning of Large Language Models”, Under Review of TACL.

Preprint

- [12] **Boyang Xue**, Qi Zhu, Hongru Wang, et al. “DAST: Difficulty-Aware Self-Training for Large Language Models”, preprint.
- [13] Rui Wang*, Hongru Wang*, **Boyang Xue***, et al. “Harnessing the Reasoning Economy: A Survey of Efficient Reasoning for Large Language Models”, preprint.

Patent

- [14] **Boyang Xue**, Cheng Cheng, Ye Yuan. “Deep Learning based Patient-Specific FHR Detection System on FECG (一种基于深度学习和特异性矫正在 FECG 上胎儿心率检测系统)”, Chinese Patent [CN112826513B].